RRA

Lab10 (Parteneriat cu laburile scrise de PETrascu)

Reglare $\mathscr{H}_2$ (o sa fac si o recaliptulare la calcularea normei $\mathscr{H}_2$ si $\mathscr{L}_2$)

Pana sa intram in reglare $\mathscr{H}_2$, mai intai o sa vreau sa vorbesc despre norma $\mathscr{L}_2$ (spatiu Lebesgue) si $\mathscr{H}_2$ (spatiu Hardy)

Pentru **Matricea de Transfer (cazul MIMO)** definim $||G||_2 := \left\{\frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{tr}\left[G^{\sim}(\mathrm{jw})G(\mathrm{jw})\right]\mathrm{dw}\right\}^{\frac{1}{2}}$ care se numeste norma $\mathscr{L}_2$

$$||G||_2 < \infty \Leftrightarrow G \in \mathscr{RL}_2(s)^{\mathrm{pxm}} \Leftrightarrow \begin{cases} G(s) \text{ nu are poli pe } \mathbb{C}_0 \\ G(\infty) = 0, \text{adica este strict proprie} \end{cases}$$

S-ar putea rezolva norma 2 folosind **Teoria Reziduurilor**, cel mai bine folosim unele formule pe stare (Lyapunov)

Presupunem ca sistemul este luat aleator si este reprezentat intr-o stare minimala. Norma lui $\mathscr{L}_2$ este finita $\Leftrightarrow \begin{cases} D = 0 \\ A \text{ este dihotomica} \end{cases}$. In acest sens, sistemul se poate aduce la forma $G(s) = G_s(s) + G_a(s)$, unde $\begin{cases} G_s(s) \text{ are polii in } \mathbb{C}_- \text{ (este stabil) sau } G_s(s) \in \mathscr{RH}_2 \\ G_a(s) \text{ are polii in } \mathbb{C}_+ \text{(este antistabil) sau } G_a(s) \in \mathscr{RH}_2^{\perp} \end{cases}$. Atunci are loc rezultatul: $||G||_2^2 = ||G_s||_2^2 + ||G_a||_2^2$.

O mica observatie: Stim ca G_s si G_a sunt ortogonale si ca $||\widetilde{G}||_2 = ||G||_2$. In acest sens, daca G_a este antistabil, asta inseamna ca $\widetilde{G_a}(s) = G_a^T(-s)$ care este stabil. Asadar, putem rescrie egalitatea in felul irmator: $||G||_2^2 = ||G_s||_2^2 + ||\widetilde{G}_a||_2^2$ si este suma de 2 stabile

Acum trebuie calculate cele 2 norme de tip $\mathscr{H}_2$. Cum facem asta? Uite aici:

Pentru A stabil, norma 2 se calculeaza:

$$||G||_2 = [\mathrm{tr}(B^T \mathrm{QB})]^{\frac{1}{2}} = [\mathrm{tr}(\mathrm{CPC}^T)]^{\frac{1}{2}}, \text{unde } P \text{ si } Q \text{ sunt solutiile unice ale ecuatiilor Lyapunov}$$

$$\begin{cases} AP + PA^T + BB^T = 0 \\ A^T Q + QA + C^T C = 0 \end{cases}$$

Pentru A instabil, norma 2 se calculeaza:

$||G||_2 = \left[\mathrm{tr}\left(B^T \widehat{Q} B\right)\right]^{\frac{1}{2}} = \left[\mathrm{tr}\left(C \widehat{P} C^T\right)\right]^{\frac{1}{2}}$, unde $\widehat{P}$ si $\widehat{Q}$ sunt solutiile unice ale ecuatiilor Lyapunov

$\begin{cases} A\widehat{P} + \widehat{P}A^T - BB^T = 0 \\ A^T \widehat{Q} + \widehat{Q}A - C^T C = 0 \end{cases}$, unde tr(A) reprezinta suma elementelor de pe diagonala principala

Daca $\widehat{P} \geq 0$, atunci perechea (A, B) este controlabila

Daca $\widehat{Q} \geq 0$, atunci perechea (C, A) este observabila

Acum intram in reglare $\mathscr{H}_2$

Fie sistemul generalizat $\begin{cases} \dot{x} = Ax + B_1 u_1 + B_2 u_2 \\ y_1 = C_1 x + D_{11} u_1 + D_{12} u_2 \\ y_2 = C_2 x + D_{21} u_1 + D_{22} u_2 \end{cases}$

Reprezentarea pe stare arata asa: $T = \begin{bmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{bmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix}$

$$\mathbf{T} = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & O & D_{12} \\ C_2 & D_{21} & O \end{array}\right] = \begin{bmatrix} \mathbf{T}_{11} & \mathbf{T}_{12} \\ \mathbf{T}_{21} & \mathbf{T}_{22} \end{bmatrix},$$

Problema optimala $\mathscr{H}_2$ arata asa: $\min\limits_{K \text{ stabil}} ||TLFi(T, K)||_2$, unde $TLFI(T, K) = T_{cl}$ (In slide-urile de curs este notat $T_{y_1 u_1}$)

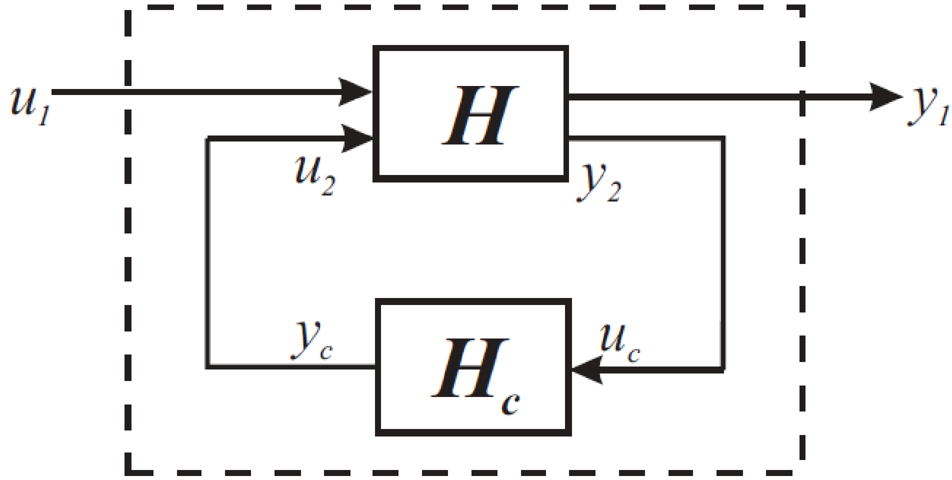


Figura 5: Transformare liniar fracționară inferioară

$TLFi(T, K) = T_{11} + T_{12}K(I - T_{22}K)^{-1}T_{21}$

Fie $K = \begin{bmatrix} A_K & B_K \\ C_K & D_k \end{bmatrix}$. Trebuie sa existe $(I - T_{22}K)^{-1}$. Pentru a asigura buna definire, trebuie ca $\begin{bmatrix} I & -D_K \\ -D_{22} & I \end{bmatrix}$ sa fie inversabil

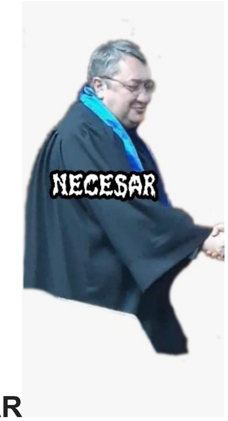
Sa zicem reprezentarea pe stare $TLFi(T, K) = T_{cl} = \begin{bmatrix} A_{\text{cl}} & B_{\text{cl}} \\ C_{\text{cl}} & D_{\text{cl}} \end{bmatrix}$

Ideea de baza este sa gasim un regulator K astfel incat $\Lambda(A_{cl}) \in \mathbb{C}_-$ (daca este satisfacuta asta, este stabilziator) si sa aiba $||T_{cl}||_2^2 < \infty$, adica $\begin{cases} T_{\text{cl}}(\infty) = 0 \\ T_{\text{cl}}(s) \text{ sa NU aiba poli in } \mathbb{C}_0 \end{cases}$. Sa detaliez la $T_{\text{cl}}(\infty) = 0$, practic $D_{\text{cl}} = 0$. Asta insemna ca $D_{11} + D_{12}D_K(I - D_{22}D_K)^{-1}D_{21} = 0$

Ce presupuneri facem:

- (A, B_2) stabilizabil
- (C_2, A) detectabil
- $D_{11} = 0$ pentru a asigura ca avem mereu solutie
- $D_K = 0$ pentru a avea buna definire asigurata a TLFi-ului si pentru a avea solutie EMAR
- $D_{22} = 0$. De asta nu este neaparat nevoie, dar se poate presupune pentru a usura calculele. Asta zice ca TLFi-u este strict propriu

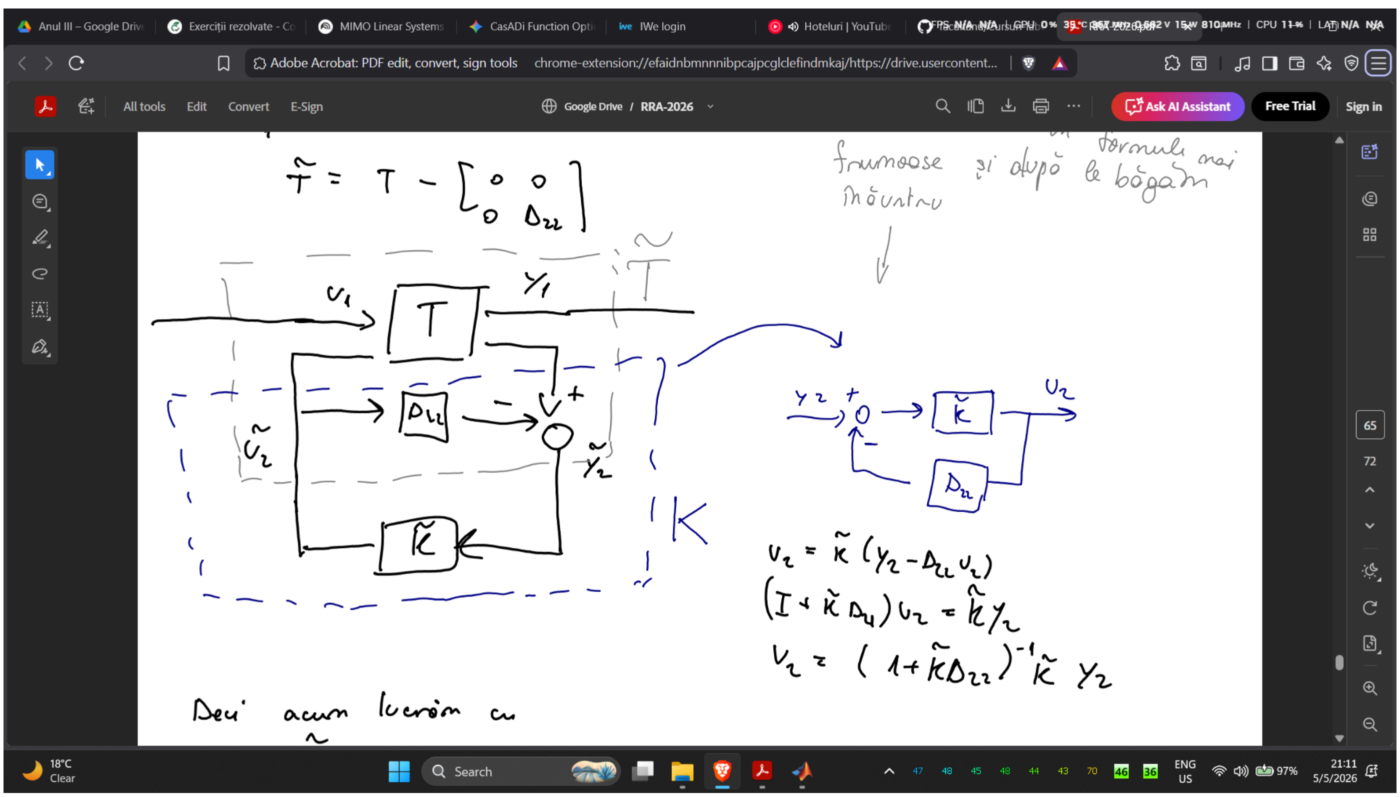
Prima si a treia presupunere ne zic ca norma 2 este finita si ca $T_{cl}(\infty) = D_{cl} = 0$. Din pacate, aceste



cerinte sunt DOAR **pentru a asigura solutia problemei $\mathscr{H}_2$**

Care-i faza cu $D_{22} = 0$:

Sa zicem ca avem $\widetilde{T} = T - \begin{bmatrix} 0 & 0 \\ 0 & D_{22} \end{bmatrix}$



De aici scoatem ca $u_2 = \widetilde{K}(y_2 - D_{22}u_2)$. Vedem ca $u_2 = \left(I + \widetilde{K}D_{22}\right)^{-1}\widetilde{K}y_2$, cu $\widetilde{K}D_{22}$ este strict propriu.

$\left(I + \widetilde{K}D_{22}\right)^{-1}\widetilde{K} = K$. Ideea e ca ne reducem mult calculele aici

O sa reatasez T pentru a ne ajuta mai departe: $T = \begin{bmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{bmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix}$

$$\mathbf{T} = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & O & D_{12} \\ C_2 & D_{21} & O \end{array}\right] = \begin{bmatrix} \mathbf{T}_{11} & \mathbf{T}_{12} \\ \mathbf{T}_{21} & \mathbf{T}_{22} \end{bmatrix},$$

Fie sistemu nostru T cu 2 triplete Popov asociate: $\begin{cases} \Sigma_{12} = \left(A; B_2; \begin{bmatrix} C_1^T C_1 & C_1^T D_{12} \\ D_{12}^T C_1 & D_{12}^T D_{12} \end{bmatrix} \right) \\ \Sigma_{21} = \left(A^T; C_2^T; \begin{bmatrix} B_1 B_1^T & B_1 D_{21}^T \\ D_{21} B_1^T & D_{21} D_{21}^T \end{bmatrix} \right) \end{cases}$. *Practic astea sunt factorilzarile inner-outer pentru T_{12} si co-inner-outer pentru T_{21}*. La Σ_{12} are sens forma. La Σ_{21} pare mai ciudatica. De ce? Noi stim 2 chestii: $\begin{cases} (C_2, A) \text{ detectabil} \Rightarrow \left(A^T, C_2^T\right) \text{ stabilizabil} \\ \Sigma = \left(A; B; \begin{bmatrix} Q & L \\ L^T & R \end{bmatrix} \right), \text{unde } Q = C^T C, R = D^T D \text{ si } L = C^T D \end{cases}$. Practic la tripletul Popov Σ_{21}, $A = A^T$ si $B = C_2^T$. Asta inseamna ca $C = B_1^T$ si $D = D_{21}^T$ si se inlocuiesc in matrice.

Pornind de la cele 2 triplete Popov, facem urmatoarele presupuneri:

H1) Stim deja ca (A, B_2) stabilizabil. Totodata, mai presupunem ca $R_{12} = D_{12}^T D_{12} > 0$ are rang integ pe coloane si ca matricea $\begin{bmatrix} A - \text{jwI} & B_2 \\ C_1 & D_{12} \end{bmatrix}$ are rang intreg pe coloane, adica $\text{rang} \begin{bmatrix} A - \text{jwI} & B_2 \\ C_1 & D_{12} \end{bmatrix} = n + m_2,\ \forall w \in \mathbb{R}$ (coloane aici). Prin asta se indeplinesc toate ipotezele de factorizare inner-outer.

H2) Stim deja ca (C_2, A) detectabil Totodata, mai presupunem ca $R_{21} = D_{21} D_{21}^T > 0$ are rang integ pe linii si ca matricea $\begin{bmatrix} A - \text{jwI} & B_1 \\ C_2 & D_{21} \end{bmatrix}$ are rang intreg pe linii, adica $\text{rang} \begin{bmatrix} A - \text{jwI} & B_1 \\ C_2 & D_{21} \end{bmatrix} = n + p_2,\ \forall w \in \mathbb{R}$ (linii aici)

Daca **H1) si H2)** au loc, atunci

- $\text{EMAR}(\Sigma_{12}) = A^T X + \text{XA} + Q - (\text{XB} + L) R^{-1} (B^T X + L^T)$ are solutie stabilizatoare $X_s \geq 0$:
  $A^T X_s + X_s A + C^T C - \left(X_s B + C_1^T D_{12}\right)\left(D_{12}^T D_{12}\right)^{-1}\left(B_2^T X_s + D_{12}^T C\right) = 0$ si reacite stabilizatoare
  $F_s := -R^{-1}(B^T X + L^T) = -\left(D_{12}^T D_{12}\right)^{-1}\left(B_2^T X_s + D_{12}^T C\right)$

- $\mathrm{EMAR}(\Sigma_{21}) = A^T X + \mathrm{XA} + Q - (\mathrm{XB} + L)R^{-1}(B^T X + L^T)$ are solutie stabilizatoare $Y_s \geq 0$:
  $A^T Y_s + Y_s A + B_1 B_1^T - \left(Y_s C_2^T + B_1 D_{21}^T\right)\left(D_{21}D_{21}^T\right)^{-1}\left(C_2 Y_s + D_{21}B_1^T\right) = 0$ si reactie stabilizatoare
  $K_s := -R^{-1}(B^T X + L^T) = -\left(D_{21}D_{21}^T\right)\left(C_2 Y_s + D_{21}B_1^T\right)$
- Regulatorul optimal si strict propriu in sens $\mathscr{H}_2$ este dat de $K_{\mathrm{opt}} = \begin{bmatrix} A + B_2 F_s + K_s^T C_2 & -K_s^T \\ F_s & 0 \end{bmatrix}$
  $K(s) = \begin{bmatrix} A + \mathrm{BF} + LC + \mathrm{LDF} & -L \\ F & 0 \end{bmatrix}$, unde $\begin{cases} \Lambda(A + \mathrm{BF}) \subset \mathbb{C}^- \\ \Lambda(A + \mathrm{LC}) \subset \mathbb{C}^- \end{cases}$. D_{22} si D_{11} si D_K am zis ca sunt 0. ***Chestia asta merge DOAR DACA*** $D_{22} = 0$. Daca am $D_{22} \neq 0$, atunci fac bucla de reactie dintre $K_{\mathrm{opt}}(s)$ si D_{22}

Valoarea optimala (minimala) a normei $\mathscr{H}_2$ este:

$\min\limits_{K \text{ stabil}} ||TLFi(T, K)||_2 = \min\limits_{K \text{ stabil}} ||T_{cl}||_2 = [tr(B_1^T X_s B_1) + tr(W_x Y_s W_x^T)]^{\frac{1}{2}} = [tr(W_y X_s W_y^T) + (C_1 Y_s C_1^T)]^{\frac{1}{2}}$. Ok, dar cine sunt $\begin{cases} W_x \\ W_y \end{cases}$? Ele ies din Sistemul Kalman-Yakubovich-Popov(SKYP). Aici o sa presupunem ca $J = I$. Avem 2 SKYP-uri, unul pentru Σ_{12} ai altul pentru Σ_{21}.

$$\begin{aligned} R &= V^T J V, \\ L + XB &= W^T J V, \\ Q + A^T X + XA &= W^T J W, \end{aligned}$$

Am zis de la inceput ca luam $J = I$. Deci putem rescrie SKYP asa: $\begin{cases} R = V^T V \\ L + \mathrm{XB} = W^T V \\ A^T X + \mathrm{XA} + Q = W^T W \end{cases}$

- Acum sa luam $\mathrm{SKYP}(\Sigma_{12}, I)$. Avem $\begin{cases} D_{12}^T D_{12} = V_x^T V_x \\ C_1^T D_{12} + X_s B_2 = W_x^T V_x \\ C_1^T C_1 + A^T X_s + X_s A = W_x^T W_x \end{cases}$. Scoatem pe V_x ca fiind factor cholesky din $D_{12}^T D_{12}$. De asemenea, noi stim cat este F_s. Totodata, stim asa: $F = -V^{-1}W$, unde F este rectia stabilizatoare din $\mathrm{EMAR}(\Sigma)$. Asadar, putem scrie ca $F_s = -V_x^{-1} W_s$. Noi stim pe V_x si F_s. Scoatem pe $W_s = -V_x F_s$

- Acum sa luam $\mathrm{SKYP}(\Sigma_{21}, I)$. Avem $\begin{cases} D_{21}D_{21}^T = V_y^T V_y \\ B_1 D_{21}^T + Y_s C_2^T = W_y^T V_y \\ B_1 B_1^T + A^T Y_s + Y_s A = W_y^T W_y \end{cases}$. Scoatem pe V_y ca fiind factor cholesky din $D_{21}D_{21}^T$. De asemenea, noi stim cat este K_s. Totodata, stim asa: $F = -V^{-1}W$, unde F este rectia stabilizatoare din $\mathrm{EMAR}(\Sigma)$. Asadar, putem scrie ca $K_s = -V_y^{-1}W_y$. Noi stim pe V_y si K_s. Scoatem pe $W_y = -V_y K_s$

Laborator

Remarca: matricile sunt alese aleator, deci la executarea scriptului o sa dea rezultate diferite

```
clear
close all
clc

A = randn(5); % sistem cu 5 stari
B2 = randn(5,2); % m2 = 2
C2 = randn(2,5); % p2 = 2
D22 = randn(2,2); % Am zis ca nu e neaparat sa il iau 0 mereu. Vrem 0 pentru
usurinta de calcul

B1 = randn(5,3); % m1 = 3
C1 = randn(3,5); % p1 = 3

D11 = zeros(3,3);
D12 = randn(3,2);
D21 = randn(2,3);
```

Acum ca ne-am definit sistemul, sa ne apucam sa rezolvam problema $\mathscr{H}_2$. Mai intai, trebuie sa vedem ca matricele D_{12} are rang integ pe coloane, respectiv D_{21} are rang intreg pe linii. *Probema:* nu am mereu garantie la matrici aleatoare ca o sa imi iasa cu rang intreg pe linii/coloane

```
D21
```

```
D21 = 2×3
   -0.4189    0.8998    1.0294
   -0.1403   -0.3001   -0.3451
```

```
rank(D21) % se poate vedea ca matricea are rang intreg pe linii
```

```
ans =
2
```

```
D12
```

```
D12 = 3×2
   -0.6086   -1.3429
   -1.2226   -1.0322
    0.3165    1.3312
```

```
rank(D12) % se poate vedea ca am rang intreg pe coloane
```

```
ans = 
2
```

Acum verific daca (A, B_2) este stabilizabil si ca (C_2, A) este detectabil. Cel mai ok ne uitam la zerourile de transmisie (bine, o sa vedem ca sistemul (A, B_2, C_2, D_{22}) este minimal, deci zerouri de transmisie = zerouri de invarianta)

```
tzero(A,B2,[],[]) % controlabil, deci si stabilizabil
```

```
ans =

  0×1 empty double column vector
```

```
tzero(A,[],C2,[]) % observabil, deci si detectabil
```

```
ans =

  0×1 empty double column vector
```

Acum trebuie sa vedem ca rangul matricilor $\text{rang}\begin{bmatrix} A - \text{jwI} & B_2 \\ C_1 & D_{12} \end{bmatrix} = n + m_2,\ \forall w \in \mathbb{R}$ si $\text{rang}\begin{bmatrix} A - \text{jwI} & B_1 \\ C_2 & D_{21} \end{bmatrix} = n + p_2,\ \forall w \in \mathbb{R}$. Daca sunt goale, atunci nu inseamna neaparat ca au rang intreg pe coloane

```
tzero(A,B2,C1,D12) % Cu asta ne asiguram ca nu pica rangul matricii asteia
```

```
ans =

  0×1 empty double column vector
```

```
tzero(A,B1,C2,D21) % La fel, sa nu pice rangul undeva
```

```
ans =

  0×1 empty double column vector
```

Acum verificam rangul matricilor si ne asiguram ca $\text{rang}\begin{bmatrix} A - \text{jwI} & B_2 \\ C_1 & D_{12} \end{bmatrix} = n + m_2,\ \forall w \in \mathbb{R}$ si

$\text{rang}\begin{bmatrix} A - \text{jwI} & B_1 \\ C_2 & D_{21} \end{bmatrix} = n + p_2,\ \forall w \in \mathbb{R}$

```
rank([A,B2;C1,D12]) % 5 stari si m2 = 2
```

```
ans = 
7
```

```
rank([A,B1;C2,D21]) % 5 stari si p2 = 2
```

```
ans =
7
```

Cu astea tin ipotezele. Adica putem rezolva cu EMAR

Mai intati ne uitam la tripletul Popov $\Sigma_{12} = \left(A; B_2; \begin{bmatrix} C_1^T C_1 & C_1^T D_{12} \\ D_{12}^T C_1 & D_{12}^T D_{12} \end{bmatrix} \right)$

```
Q12 = C1' * C1
```

```
Q12 = 5×5
    5.2224   -4.0024   -1.2169    0.9295    0.6999
   -4.0024    7.8843    2.2321   -3.1544   -0.8119
   -1.2169    2.2321    3.3753   -4.1109   -0.4140
    0.9295   -3.1544   -4.1109    5.2224    0.4726
    0.6999   -0.8119   -0.4140    0.4726    0.1209
```

```
L12 = C1' * D12
```

```
L12 = 5×2
   -2.2442   -1.2161
    3.9138    5.3278
    1.0467    2.4433
   -1.4314   -3.5947
   -0.4219   -0.4771
```

```
R12 = D12' * D12
```

```
R12 = 2×2
    1.9653    2.5005
    2.5005    4.6408
```

Acum vedem tripletul Popov $\Sigma_{21} = \left(A^T; C_2^T; \begin{bmatrix} B_1 B_1^T & B_1 D_{21}^T \\ D_{21} B_1^T & D_{21} D_{21}^T \end{bmatrix} \right)$

```
Q21 = B1 * B1'
```

```
Q21 = 5×5
    7.9910   -3.1254   -5.0530   -3.3334   -3.1976
   -3.1254    1.4740    2.2330    1.3675    1.2395
   -5.0530    2.2330    3.4989    2.0466    1.9538
   -3.3334    1.3675    2.0466    1.7878    1.5026
   -3.1976    1.2395    1.9538    1.5026    1.3573
```

```
L21 = B1 * D21'
```

```
L21 = 5×2
    1.9848   -0.7127
   -1.3450    0.3675
   -1.9408    0.6161
   -0.6535    0.0938
   -0.6257    0.1795
```

```
R21 = D21 * D21'
```

```
R21 = 2×2
    2.0448   -0.5665
   -0.5665    0.2288
```

Acum aflam de la $\Sigma_{12} = \left(A; B_2; \begin{bmatrix} C_1^T C_1 & C_1^T D_{12} \\ D_{12}^T C_1 & D_{12}^T D_{12} \end{bmatrix} \right)$ X_s si F_s

```
[Xs,Fs,spectrus12] = icare(A,B2,Q12,R12,L12)
```

```
Xs = 5×5
   16.7987  -12.4916    1.5169  -14.4783  -24.0990
  -12.4916   11.5903   -3.6821    7.0676   12.1485
    1.5169   -3.6821    3.4720    2.8363    0.9745
  -14.4783    7.0676    2.8363   28.3900   29.2237
  -24.0990   12.1485    0.9745   29.2237   70.3221
Fs = 2×5
    6.6231   -7.7418    3.4882  -10.7247   -7.9218
   -4.3428    6.3636   -1.9868    5.1238    2.2252
spectrus12 = 5×1 complex
  -0.4545 + 1.4504i
  -0.4545 - 1.4504i
  -0.6478 + 0.0000i
  -2.3205 + 0.0000i
  -4.5839 + 0.0000i
```

```
Fs = -Fs; % chestia asta pentru reactie pozitiva
eig(A + B2*Fs)
```

```
ans = 5×1 complex
  -4.5839 + 0.0000i
  -2.3205 + 0.0000i
  -0.6478 + 0.0000i
  -0.4545 + 1.4504i
  -0.4545 - 1.4504i
```

```
eig(Xs) % verificam ca solutia Riccati este pozitiv semi-definita
```

```
ans = 5×1
    0.0004
    1.8203
   13.6980
   16.2006
   98.8536
```

Acum aflam de la $\Sigma_{21} = \left(A^T; C_2^T; \begin{bmatrix} B_1 B_1^T & B_1 D_{21}^T \\ D_{21} B_1^T & D_{21} D_{21}^T \end{bmatrix} \right)$ pe Y_s si K_s

```
[Ys,Ks,spectrus21] = icare(A',C2',Q21,R21,L21)
```

```
Ys = 5×5
    2.3484   -0.3441   -1.2594   -1.0471   -0.0069
   -0.3441    8.9923   -2.6121    9.3430   -0.4145
   -1.2594   -2.6121    1.7586   -2.0742    0.2039
   -1.0471    9.3430   -2.0742   10.3122   -0.4527
   -0.0069   -0.4145    0.2039   -0.4527    0.1933
Ks = 2×5
    3.1945    2.6505   -3.5493    2.5960   -1.7562
   11.2256   -4.4250   -6.9861   -8.7557   -4.2661
```

```
spectrus21 = 5×1 complex
  -0.5577 + 1.5618i
  -0.5577 - 1.5618i
  -1.3498 + 0.0000i
  -2.2865 + 0.0000i
 -11.4313 + 0.0000i
```

```
Ks = -Ks'; % negarea am zis de ce. Transpunerea este de la dualitate (recomand revedere curs TSA)
eig(A + Ks*C2) % na
```

```
ans = 5×1 complex
 -11.4313 + 0.0000i
  -0.5577 + 1.5618i
  -0.5577 - 1.5618i
  -2.2865 + 0.0000i
  -1.3498 + 0.0000i
```

```
eig(Ys)
```

```
ans = 5×1
    0.0083
    0.1835
    0.3236
    3.4077
   19.6817
```

Partea 1 din teorema merge

```
K_opt = ss(A + B2*Fs + Ks*C2, -Ks, Fs, 0) % cand D22 = 0
```

```
K_opt =
 
  A = 
            x1       x2       x3       x4       x5
   x1    1.878   -10.98   0.7181   -7.851    15.59
   x2     9.83   -7.511    1.481   -18.82   -12.33
   x3   0.7384     1.72    1.044    4.427   -12.59
   x4     6.09    1.351   0.3782   -13.98   -17.56
   x5    2.368  -0.4948    1.837    1.048   -8.514
 
  B = 
           u1      u2
   x1   3.194   11.23
   x2   2.651  -4.425
   x3  -3.549  -6.986
   x4   2.596  -8.756
   x5  -1.756  -4.266
 
  C = 
           x1      x2      x3      x4      x5
   y1  -6.623   7.742  -3.488   10.72   7.922
   y2   4.343  -6.364   1.987  -5.124  -2.225
 
  D = 
       u1  u2
   y1   0   0
   y2   0   0
 
Continuous-time state-space model.
Model Properties
```

```
K_opt_h2 = feedback(K_opt,D22) % cand D22 != 0
```

```
K_opt_h2 =

  A = 
            x1       x2       x3       x4       x5
   x1    42.91    -46.3    25.28   -93.06   -62.69
   x2    18.59   -23.14    4.835   -25.01   -10.37
   x3   -34.76     35.5   -19.45    73.34    47.66
   x4    6.527   -9.553   -1.817   0.7209    5.891
   x5   -16.69    17.02   -9.307    38.95    25.24

  B = 
            u1       u2
   x1    3.194    11.23
   x2    2.651   -4.425
   x3   -3.549   -6.986
   x4    2.596   -8.756
   x5   -1.756   -4.266

  C = 
            x1       x2       x3       x4       x5
   y1   -6.623    7.742   -3.488    10.72    7.922
   y2    4.343   -6.364    1.987   -5.124   -2.225

  D = 
        u1  u2
   y1    0   0
   y2    0   0

Continuous-time state-space model.
Model Properties
```

```
T = ss(A,[B1,B2],[C1;C2],[D11,D12;D21,D22]); % T nostru original
T_cl = lft(T,K_opt_h2)
```

```
T_cl =

  A = 
                x1         x2         x3         x4         x5         x6         x7         x8         x9        x10
   x1       0.6125     -0.993     0.1992     0.9421     0.1202      6.938     -8.877      3.475      -10.1     -6.531
   x2     -0.05489      0.975     -1.521     0.3005     0.5712      8.546     -11.34      4.186     -11.84     -7.113
   x3       -1.119    -0.6407    -0.7236    -0.3731     0.4128     -2.032       2.63     -1.011      2.912      1.842
   x4      -0.6264      1.809    -0.5933     0.8155     -0.987       3.46     -4.466      1.724     -4.977     -3.165
   x5       0.2495      -1.08     0.4013     0.7989     0.7596    -0.1618     0.4959   -0.01364    -0.1927     -0.514
   x6        5.673      1.115      2.956     -1.306        -22      1.878     -10.98     0.7181     -7.851      15.59
   x7       -1.338     -2.851      1.185      7.278       5.79       9.83     -7.511      1.481     -18.82     -12.33
   x8       -3.889     0.2694     -2.778     -1.888      14.84     0.7384       1.72      1.044      4.427     -12.59
   x9       -3.256     -4.008     0.7522      9.819      13.41       6.09      1.351     0.3782     -13.98     -17.56
   x10       -2.28   -0.08912     -1.449    -0.4418       8.76      2.368    -0.4948      1.837      1.048     -8.514

  B = 
                u1         u2         u3
   x1       0.1837      2.787    -0.4336
   x2       0.2908     -1.167    -0.1685
   x3       0.1129     -1.854    -0.2185
   x4         0.44     -1.141     0.5413
   x5       0.1017     -1.093     0.3893
   x6       -2.913    -0.4945    -0.5853
   x7      -0.4894      3.713      4.255
   x8        2.467     -1.097     -1.243
   x9       0.1411      4.964      5.694
   x10       1.334       -0.3    -0.3357
```

```
  C = 
             x1         x2         x3         x4         x5         x6         x7         x8         x9        x10
   y1    0.7512     -1.283     -1.836      2.227     0.2358     -1.801      3.834    -0.5452     0.3537     -1.833
   y2     1.778     -2.329    0.06676   -0.06921     0.2458      3.615     -2.897      2.214     -7.823     -7.388
   y3     1.223     0.9019    0.03548    -0.5073    0.07005      3.685     -6.021      1.541     -3.426    -0.4549

  D = 
       u1  u2  u3
   y1   0   0   0
   y2   0   0   0
   y3   0   0   0

Continuous-time state-space model.
Model Properties
```

```
eig(T_cl.A)
```

```
ans = 10×1 complex
 -11.4313 + 0.0000i
  -4.5839 + 0.0000i
  -0.5577 + 1.5618i
  -0.5577 - 1.5618i
  -0.4545 + 1.4504i
  -0.4545 - 1.4504i
  -2.3205 + 0.0000i
  -2.2865 + 0.0000i
  -0.6478 + 0.0000i
  -1.3498 + 0.0000i
```

```
spectrus21
```

```
spectrus21 = 5×1 complex
  -0.5577 + 1.5618i
  -0.5577 - 1.5618i
  -1.3498 + 0.0000i
  -2.2865 + 0.0000i
 -11.4313 + 0.0000i
```

```
spectrus12 % respecta principuil separatiei Kalman asta
```

```
spectrus12 = 5×1 complex
  -0.4545 + 1.4504i
  -0.4545 - 1.4504i
  -0.6478 + 0.0000i
  -2.3205 + 0.0000i
  -4.5839 + 0.0000i
```

```
norm(T_cl,2) % norma 2 a lui TLFi(T,K)
```

```
ans = 
56.2734
```

Verificam cu formula ca astea e ok

$$V_x^T V_x = D_{12}^T D_{12}.$$

```
Vx = chol(D12' * D12,'lower')'
```

```
Vx = 2×2
    1.4019    1.7837
```

```
         0    1.2080
```

```
norm(D12' * D12 - Vx' * Vx) % paranoia lui Andrei cu chol:)))
```

```
ans =
2.2204e-16
```

$V_y^T V_y = D_{21} D_{21}^T$

```
Vy = chol(D21 * D21', 'lower')'
```

```
Vy = 2×2
    1.4299   -0.3961
         0    0.2681
```

```
norm(D21 * D21' - Vy' * Vy) % paranoia lui Andrei cu chol:)))
```

```
ans =
4.4409e-16
```

Acum scot $W_x = -V_x F_s$

```
Wx = -Vx * Fs
```

```
Wx = 2×5
    1.5386    0.4975    1.3462   -5.8956   -7.1365
   -5.2462    7.6873   -2.4001    6.1896    2.6880
```

Acum scot $W_y = -V_y K_s^T$

```
Wy = -Vy * Ks' % transpun ca sa aiba sens inmultirea matriceala (e practic Ks din
EMAR netranspus)
```

```
Wy = 2×5
    0.1210    5.5430   -2.3077    7.1807   -0.8213
    3.0100   -1.1865   -1.8733   -2.3477   -1.1439
```

Acum vedem ambele formule pentru problema noastra $\mathscr{H}_2$:

$$\min_{K \text{ stabil}} ||TLFi(T,K)||_2 = \min_{K \text{ stabil}} ||T_{cl}||_2 = [tr(B_1^T X_s B_1) + tr(W_x Y_s W_x^T)]^{\frac{1}{2}} = [tr(W_y X_s W_y^T) + (C_1 Y_s C_1^T)]^{\frac{1}{2}}$$

```
norma2_manual_1 = sqrt(trace(B1' * Xs * B1) + trace(Wx * Ys * Wx'))
```

```
norma2_manual_1 =
56.2734
```

```
norma2_manual_2 = sqrt(trace(Wy * Xs * Wy') + trace(C1 * Ys * C1'))
```

```
norma2_manual_2 =
56.2734
```